

Balaji P
Sr Data Engineer
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PROFESSIONAL SUMMARY:

- Extensive AWS Experience Deep specialization in AWS Glue, Redshift, Athena, Amazon S3, and migration from on-premises data systems to AWS. Proven expertise in developing data engineering solutions for large-scale data pipelines and ETL workflows.
- AWS Glue & Redshift Developed and optimized AWS Glue ETL jobs for data transformation, cleansing, and incremental data loading into Redshift and Hudi. Leveraged Glue Crawlers for automatic schema detection and metadata creation in the Glue Catalog.
- Migration Expertise Successfully designed and implemented data migration strategies to transfer data from on-premises systems (Oracle, SQL Server, DB2, MongoDB) to AWS using Glue, Redshift, and S3. Managed schema mapping, data integrity checks, and performance tuning during the migration.
- Automated Workflow Development Automated 20+ complex data pipelines using AWS Glue Workflows and Triggers, reducing manual reporting efforts by 60% and enhancing operational efficiency by 30%.
- Data Analysis & Querying Expertise in Redshift SQL, Athena, and Spark-SQL for data migration, manipulating and reporting. Proficient in optimizing queries for improved performance in large-scale datasets.
- Data Warehousing Proficiency Skilled in developing and maintaining robust Data Warehousing models for data accuracy, consistency, and scalability.
- Cost Optimization Implemented strategies that reduced operational costs by 30%, leveraging Glue Bookmarks for incremental processing, optimized Redshift distribution keys, and efficient data storage in Amazon S3.
- Python Utilities Experience in creating Python scripts and utilities to automate repetitive tasks, enhance the functionality of the framework, and support data validation and error handling.
- Designed and automated 20+ data pipelines using AWS Glue, Airflow, and Step Functions, enhancing data availability and reducing manual intervention by 60%.
- Migrated petabyte-scale data from on-premises Oracle, SQL Server, and DB2 to AWS Redshift, ensuring data accuracy, schema mapping, and performance optimization.
- Optimized Spark-based ETL workflows on AWS EMR and Scala, reducing data latency by up to 30% and improving AI model integration.
- Implemented Docker-based deployments for Spark/Scala jobs, cutting down environment setup time and enhancing CI/CD efficiency.

TECHNICAL SKILLS:

Programming Languages	Python, Scala, SQL
Data Warehousing	Snowflake, SQL Datawarehouse, Redshift
ETL Tools	Apache NiFi, Apache Airflow, Informatica, SSIS, AWS Glue
Data Processing Frameworks	Apache Spark, Apache Flink, Apache Storm
Database Systems	PostgreSQL, MySQL, Oracle, Microsoft SQL Server
Big Data Technologies	Hadoop, Hive, HBase, Kafka, Oozie,
Cloud Platforms	AWS (Amazon Web Services), Azure
Data Visualization	Power BI, Tableau
Version Control	Git, GitHub, Bitbucket
Data Modelling	Erwin Data Modeler, dbt (data build tool)
Web Technologies	HTML, CSS, React, Angular
Scripting Languages	UNIX Shell script, Python, R

Data-Modeling Methodologies	Object Relational Modeling, ER modeling, Dimensional Modeling
Cloud technologies	Data Lake Analytics, Databricks, Data Factory, AWS Glue, Blob Storage
Development Methodologies	Agile, Scrum, Iterative Development, Waterfall Model, UML,
Bigdata distribution	Cloudera, Hortonworks, Amazon EMR

PROFESSIONAL EXPERIENCE:

Frankcrum Insurance Agency Inc
June 2023 - Current.
Sr Data Engineer

Responsibilities:

- Upgraded Tech Stack for AI-Driven Ad Models Upgraded Scala 2.x to 3.x and Apache Spark 2.x to Spark 3.x, ensuring compatibility with AI-powered ad models.
- Conducted thorough performance tuning and dependency conflict resolution post-upgrade, leading to improved data processing efficiency.
- Migration to AWS EMR-6 for Scalability & Performance Migrated data pipelines to AWS EMR-6, leveraging optimized resource allocation for large-scale data processing.
- Enhanced pipeline scalability and reliability, ensuring high availability for ad decision-making processes.
- Engineering a Data Ingestion Framework for App Search Data Designed and implemented a data ingestion framework to capture App IDs searched by users in the Play Store.
- Efficiently stored this data in a distributed data lake, ensuring high availability and quick retrieval.
- Optimized Spark-Based ETL Pipelines for Real-Time Ad Targeting Developed high-performance Spark RDD-based ETL pipelines for efficient processing, transformation, and storage of app search data.
- Upgraded to Spark 3.x with adaptive execution, significantly improving pipeline processing speed and query performance.
- Reduced data latency by 30% through optimized Spark RDD transformations and storage strategies.
- Seamless AI Model Integration for Ad Selection Optimization Ensured real-time processing of app search data to feed AI-driven ad selection algorithms.
- Enabled more accurate ad targeting by improving data preparation and model input quality.
- Collaborated with ML engineers, data scientists to align upgraded pipeline with AI model requirements.
- Upgraded & Optimized Docker Images for Deployment Efficiency Upgraded Docker base images to ensure compatibility with Scala 3.x, Spark 3.x, and AWS EMR-6.
- Optimized Docker file by reducing image size, implementing multi-stage builds for better efficiency.
- Ensured seamless deployment by testing and validating the new Docker images across different environments.

Environment: Oozie, Apache Spark, Scala, Python, AWS (Amazon S3, Amazon EMR), Airflow, Rio (CI/CD), Kubernetes, Git, AWS (Amazon S3, Amazon EMR), Docker.

NIIT Technology, Ind 2022
Data Engineer

Jun 2021 to Aug

Responsibilities:

- Developed AWS Glue jobs and Python scripts to efficiently load data from source systems into data warehouse tables, ensuring data integrity and consistency during the loading process.

- Worked with COBOL copybooks to parse and load legacy data into Amazon S3, facilitating the migration of older data formats.
- Extracted data from various sources such as Oracle, Teradata, and SQL Server using AWS Database Migration Service (DMS) and AWS Glue, ensuring smooth data migration to Amazon Redshift.
- Created Glue Crawlers to automatically detect and infer schema, building Glue Data Catalog tables for seamless data integration.
- Developed complex SQL queries in Amazon Redshift to analyze and transform data efficiently, ensuring accurate reporting and data visualization.
- Designed and implemented ETL pipelines in AWS Glue to extract, transform, and load data from Amazon S3 into Amazon Redshift, leveraging Glue Bookmarks to manage incremental data loads and avoid duplication.
- Developed and optimized Spark-based ETL jobs using AWS Glue for batch data processing and transformation to ensure efficient data flow within the warehouse.
- Utilized AWS Lambda for triggering Glue workflows, automating data validation, pre- and post-load checks, and monitoring job status.
- Managed data ingestion pipelines using AWS Glue, AWS DMS, and AWS Step Functions to ensure automated workflow execution and error handling.
- Implemented data transformation logic using Glue PySpark jobs, converting legacy formats into optimized Parquet and Avro formats to improve performance and reduce storage costs.
- Developed Python scripts to automate data quality checks, anomaly detection, and reconciliation of migrated data between on-premises and AWS systems.
- Utilized Amazon CloudWatch to monitor Glue jobs, ensuring performance optimization and real-time error detection.
- Conducted data validation and reconciliation to ensure data completeness and accuracy in Amazon Redshift post-migration.
- Optimized Redshift performance by implementing appropriate distribution keys, sort keys, and compression encoding strategies.
- Utilized AWS Step Functions to orchestrate complex workflows, ensuring end-to-end automation of data processing and reporting pipelines.
- Designed a scalable data processing framework using AWS Glue and Redshift to facilitate automated reporting of financial data, significantly improving data accessibility for stakeholders.

Environments: AWS Glue (Jobs, Crawlers, Catalog, Bookmarks), Amazon S3, Amazon Redshift, AWS DMS, AWS Lambda, AWS Step Functions, Amazon CloudWatch, Power BI, SQL, Data Modeling, Parquet, Avro, PySpark.

**Accretive Health Services, Ind
Jun 2021
Data Engineer**

Jan 2019 to

Responsibilities:

- Designed and developed AWS Glue ETL pipelines to ingest and transform financial data from multiple sources (MySQL, Salesforce, and retail systems) into an optimized data model for Redshift reporting.
- Developed complex SQL transformations in Amazon Redshift to aggregate financial data, ensuring high-performance reporting for monthly and yearly financial summaries.
- Utilized AWS Lambda to automate data validation, encryption processes, and dynamic trigger-based workflows to notify stakeholders of report completion.
- Designed a robust data encryption strategy using AWS Key Management Service (KMS) to securely handle financial reports before transmission to Amazon S3.
- Integrated Amazon Event Bridge to trigger automated email alerts upon successful or failed data pipeline runs, ensuring proactive communication with stakeholders.
- Developed Python-based utilities for data quality checks, enabling automated data reconciliation between source systems and Redshift to identify variances.
- Leveraged AWS Glue Bookmarks to track processed data, ensuring incremental data loads and reducing redundancy.

- Utilized AWS Step Functions to orchestrate complex data workflows, ensuring dependency management, retry logic, and automated recovery from job failures.
- Developed Unix shell scripts for additional automation tasks such as log management, error tracking, and data sanitization.
- Designed comprehensive data validation frameworks to compare EDW data against source systems, ensuring data integrity across the reporting pipeline.
- Conducted performance tuning on Redshift queries using distribution keys, sort keys, and compression techniques to reduce data retrieval latency by 40%.
- Created detailed technical documentation outlining pipeline design, data flow, and troubleshooting guidelines to support operational continuity.
- Managed sprint planning sessions, conducted daily SCRUM meetings, and ensured that deliverables met quality standards and project timelines.
- Delivered training sessions for team members on Glue, Redshift, and AWS best practices, ensuring knowledge transfer and effective project handover.

Environments: AWS Glue (Jobs, Crawlers, Catalog, Bookmarks), Amazon Redshift, AWS Lambda, AWS Step Functions, Amazon Event Bridge, AWS KMS, Amazon S3, Python, Power BI, SQL, Unix Shell Scripting